
[IRL]² MakiMate



ME/ECE 4999

Group 9

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What is MakiMate?



Open Source Social Robot



Low Cost and Reproducible



ROS2 Based and Customizable



Built for Natural Interaction



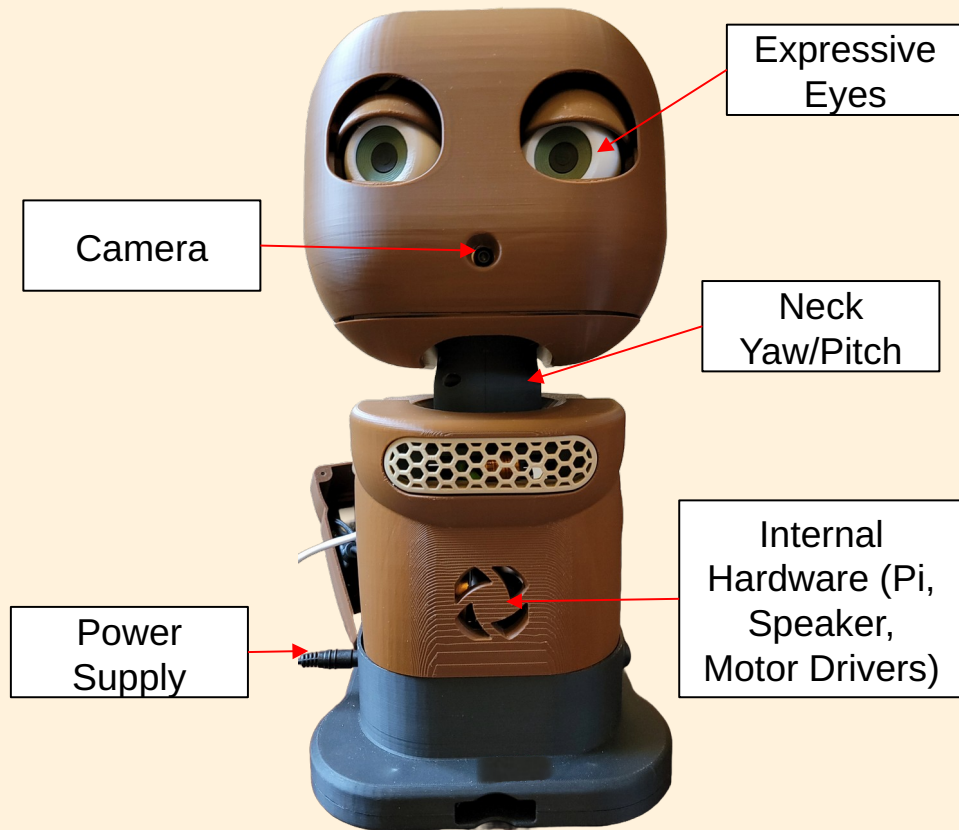
The Team's Deliverables



- Speaker identification in noisy environment
- Addition of arms and torso redesign
- Implementation of customizable LLM

System Overview

- Camera tracks user attention
- External mic and internal speaker enable dialogue
- Six servos drive expression
- Raspberry Pi coordinates behavior



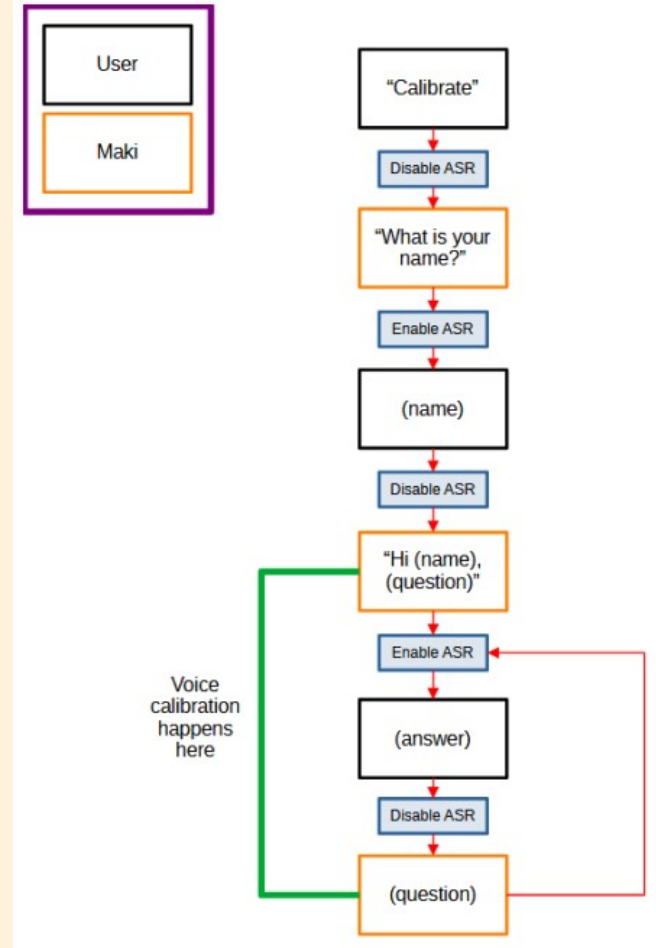
Hearing & Seeing

- 30% improved accuracy
- Layered noise filtering
- Locked gaze
 - 85% speaker turn accuracy
 - 1.2s switching latency



Personalization

- Calibration sequence
- Creates user profiles
- Social robot feature



 **MakiMate Dashboard** ConnectedFilter node... INFO+ ☒ Auto-scroll

[makimate_dashboard] Dashboard client connected (1 total)

▶ ROBOT CONTROL

● Unknown

🔊 VOLUME

Speaker Output 80%Mic Input Gain 80%

⚙️ SERVER CONFIG

LLM URL LLM Model Whisper URL Whisper Model ▼

⚙️ PARAMETERS

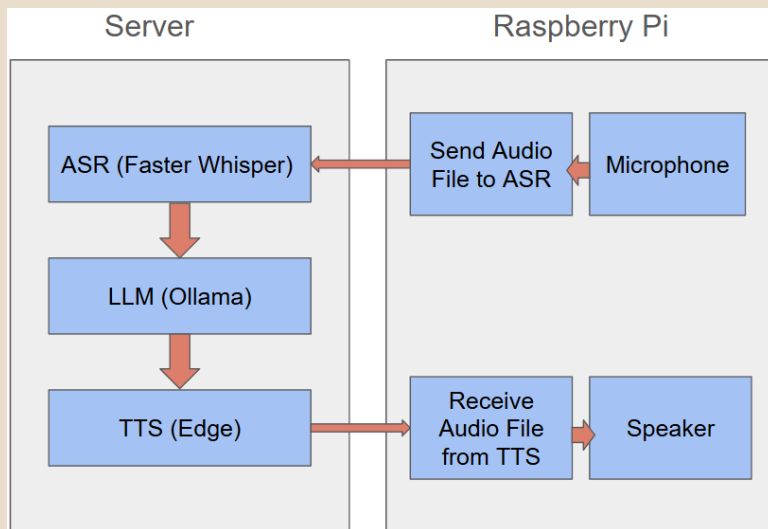
maki_dxl_6

Servo smoothing - how fast motors physically move

Smoothing Alpha 0.28 Motor Update Rate (Hz) 30.00

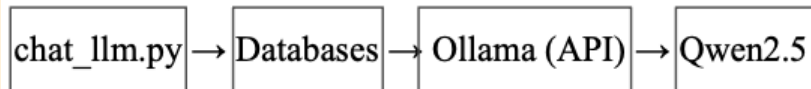
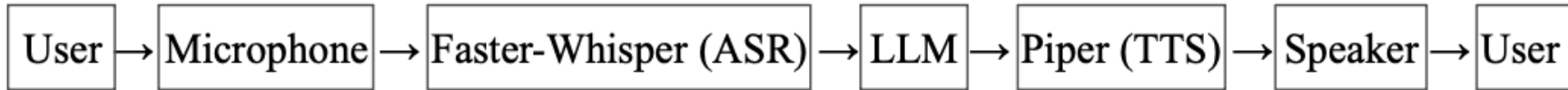
LLM driven Conversational Ability

- ASR → LLM → TTS Pipeline
- Faster Speeds
- Integrated Databases
- Reduced Cost



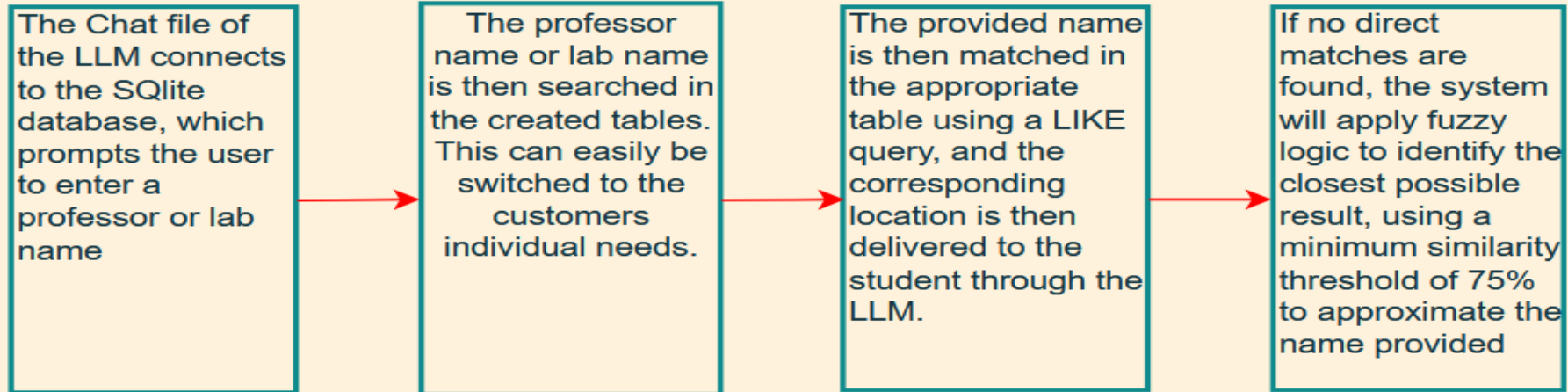
LLM Modularity

- General Socialization
- Specific Uses
 - EC Greeter: Directions, Questions
 - User Defined Prompt Potential
 - Switches Prompts at Click of Button



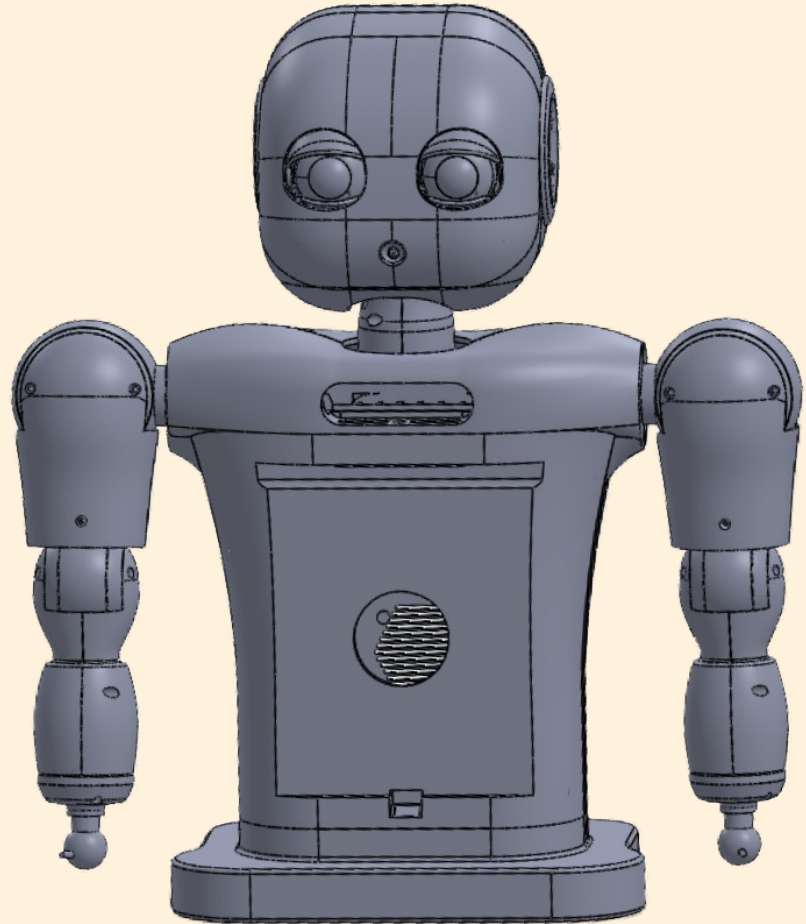
Databases

- A database stores information locally in a single file.
- After the testing of different databases, we selected SQLite.
- It was found to be 50% faster than the next closest tested.
- It was then incorporated into Main LLM file.
- To help with different accents, Fussy logic was incorporated.



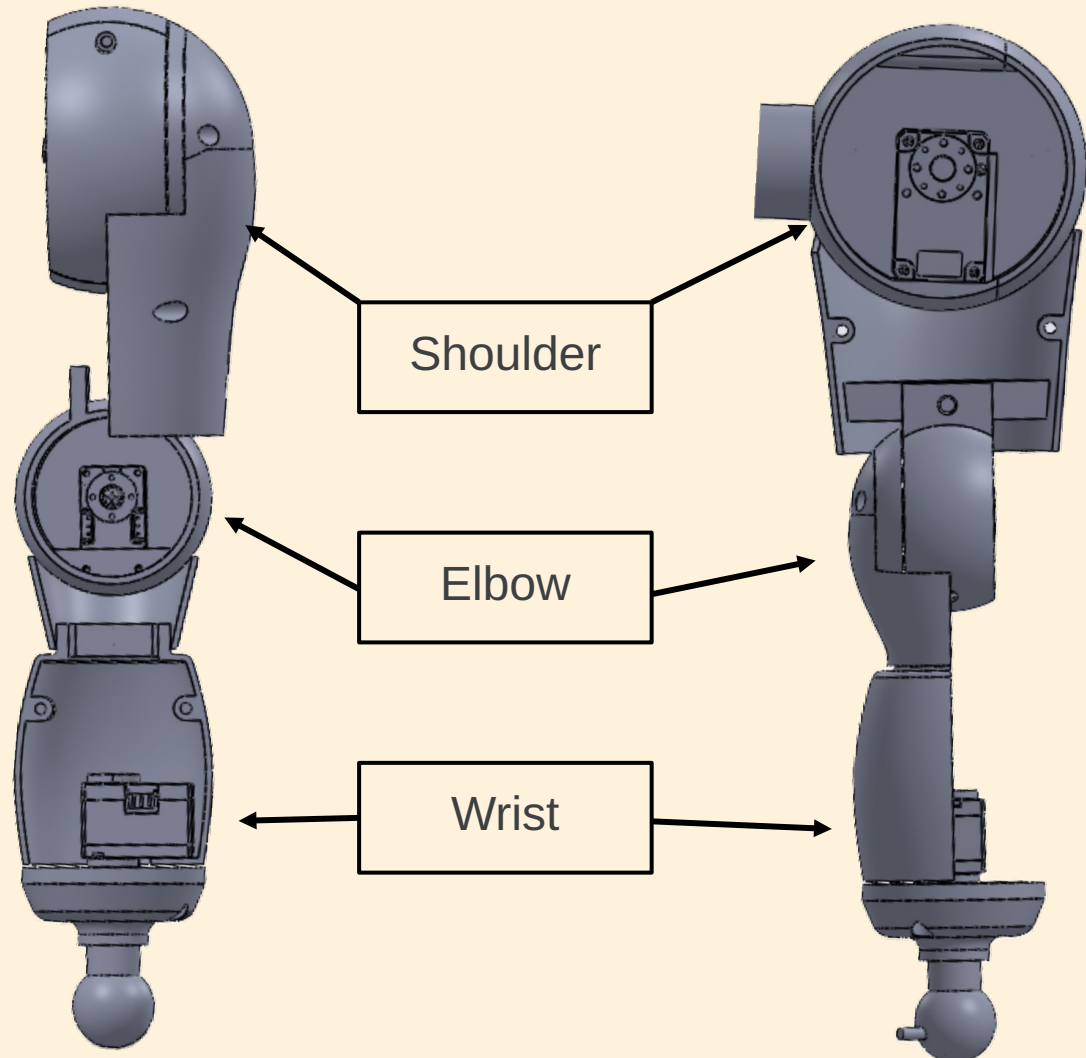
Arm Overview

- Comprehensive Design for Gesture Capability
- 4 DOF per Arm
- Bubbly Appearance



Arm Design

- Motor-Driven Revolute Joints
- Encapsulated Wires
- Easy and Reliable Assembly



Arm Results

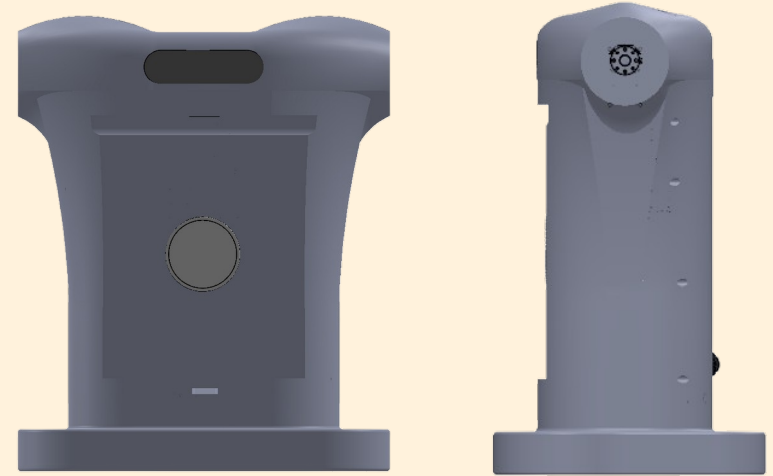
- Validated 4-DOF Functionality
- Joint ROM exceed targets
 - Shoulder 90° ($>75^\circ$)
 - Elbow 80° ($>75^\circ$)
 - Wrist 180° ($=180^\circ$)
- Servos comfortably provide required torque
 - Wrist: 0.104N ($> 0.02\text{N}$)
 - Elbow: 0.3N ($> 0.27\text{N}$)
 - Shoulder: 0.82N ($= 0.82\text{N}$)



Torso Design Evolution



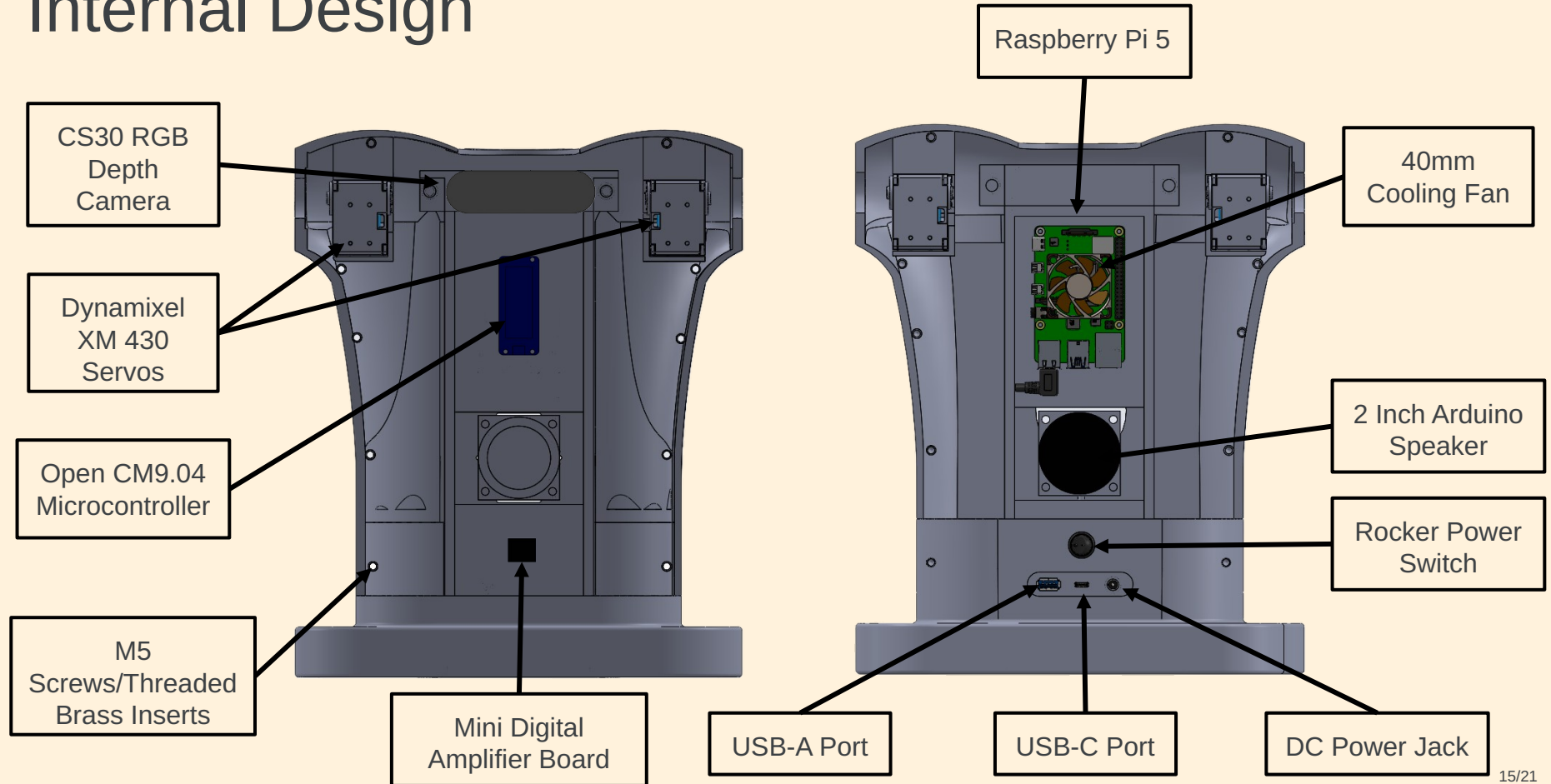
Height: 7.2 in Width: 5.4 in Depth: 4 in



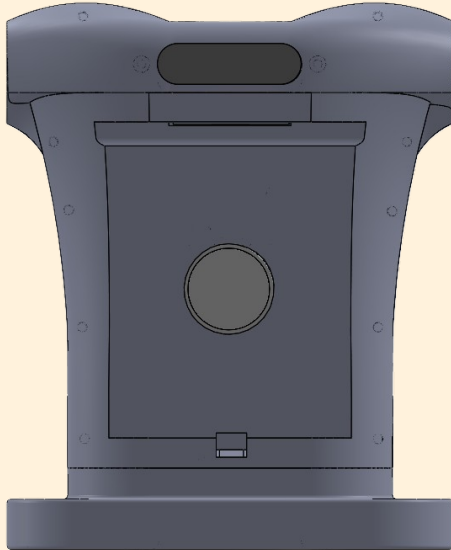
Height: 13 in Width: 8 in Depth: 5 in

- Dimensions meet “baby schema” proportion criteria
- Supports tablet integration & future hardware expansion
- Supports Arm Integration

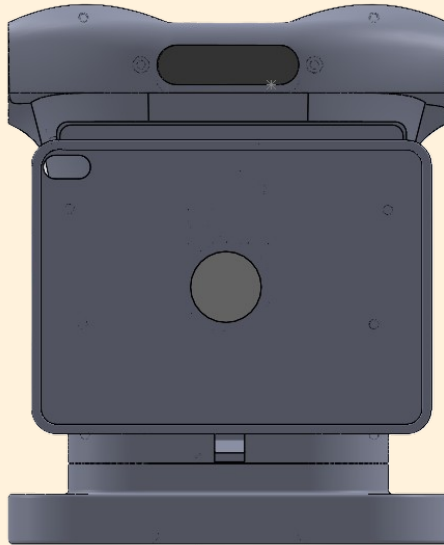
Internal Design



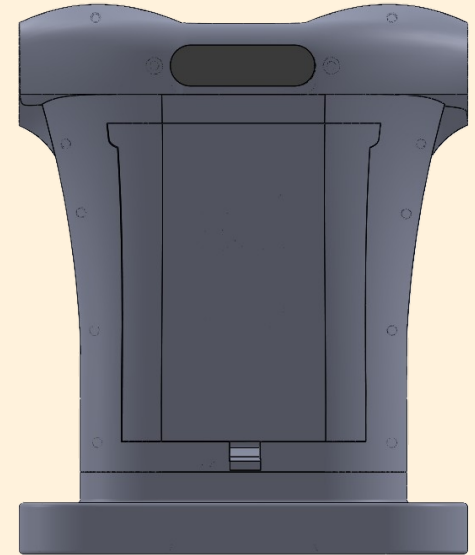
Tablet Integration



- Spring clip retention for quick and tool-less swapping

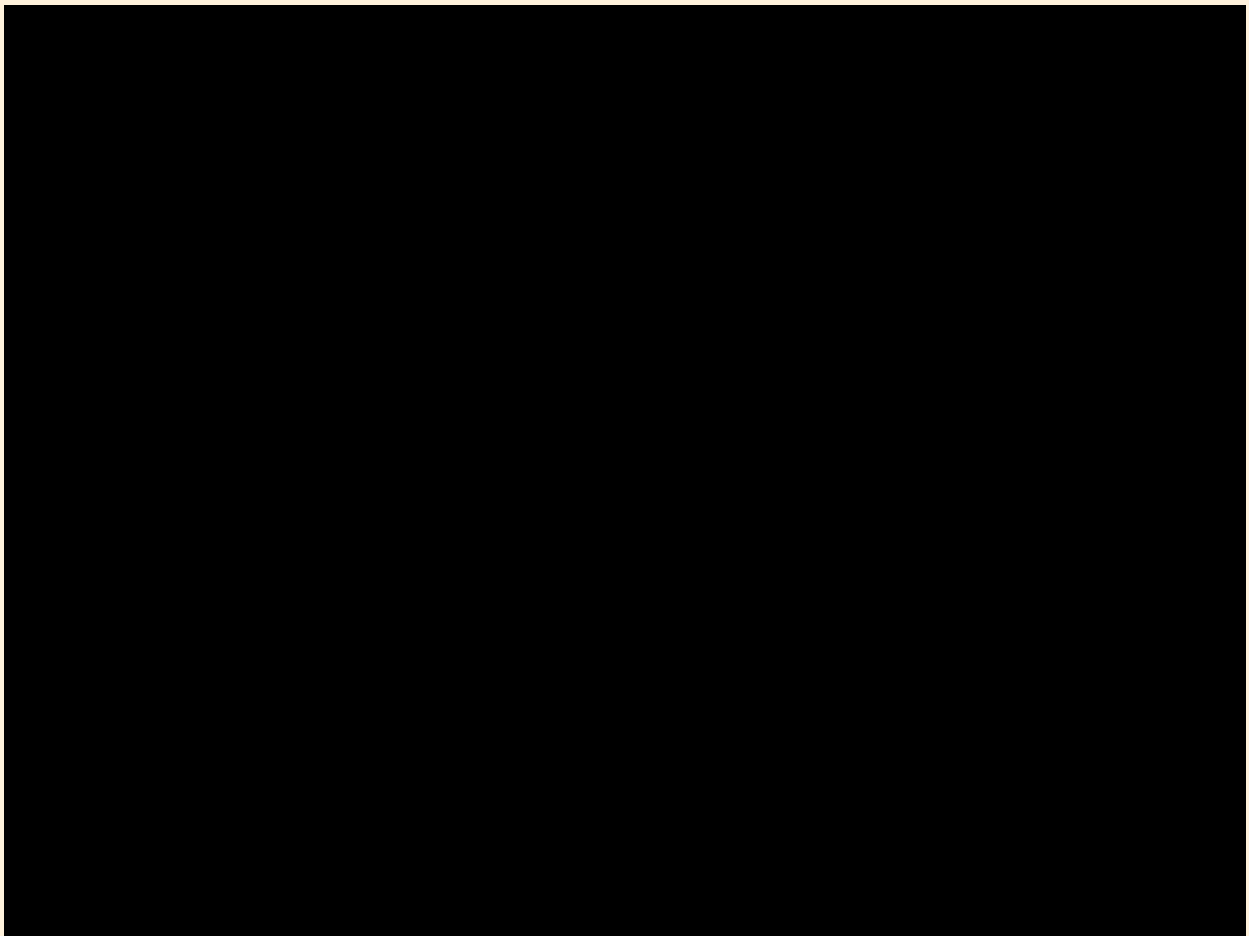


- Removable tablet holder with integrated Mag-Safe charging



- Interchangeable cover panel for smooth exterior finish





Demo Hardware

Last Semester

- 12th Gen Intel i9
- RTX 5050

This Semester

- Ryzen 5 5500u
- No dedicated GPU

Thank you!